Urban Analytics & Spatial Data Portfolio

JINYANG XU

better urban analytics

Please visit full projects through this link :)

Or scroll down for a brief introduction to each project.

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Portfolio Overview

This portfolio highlights projects that combine big data processing, remote sensing, deep learning, spatial analysis, statistical modeling, and planning interpretation. It is focusing on turning socioeconomic, demographic, real estate, fiscal, and spatial data into clear research findings and practical recommendations.

Selected Projects

• Interactive Mapping Platform for Rental Subsidy and Neighborhood Context

• Philadelphia Housing Price Modeling

• Satellite Deep Learning for Habitat Land-Cover Segmentation

• Modeling Crime Risk Around SEPTA Bus Stops

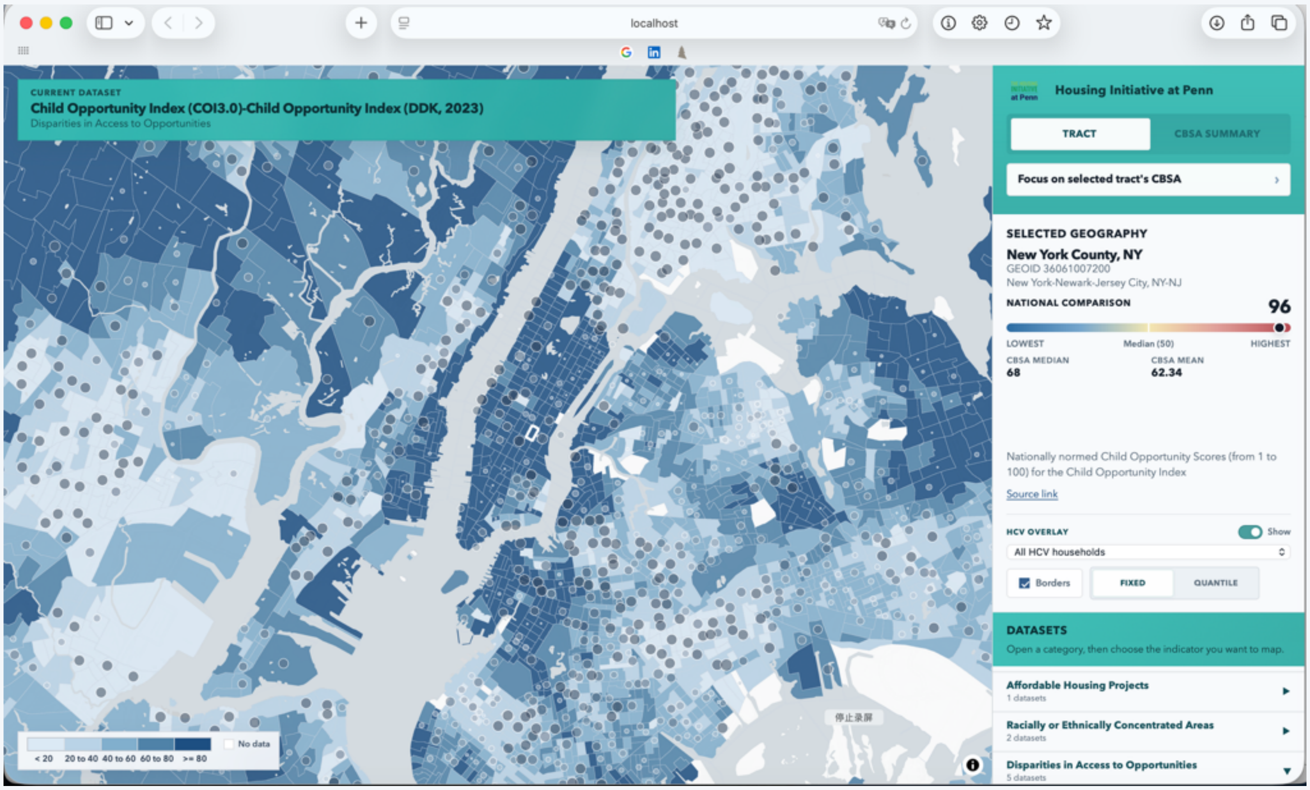
• Individual Activity Pattern Analysis Using Mobile Signal Data

• Cuisine Influence Index (CII): Philadelphia Food Culture Diffusion

• LLM-Based Street-View Walkability Analysis

1. Interactive Mapping Platform for Rental Subsidy and Neighborhood Context

Housing, affordability, neighborhood opportunity, and interactive mapping



map/dashboard screenshot from rental subsidy platform

Project Focus: Developed an interactive U.S. national platform to visualize rental subsidy, housing affordability, and neighborhood context indicators.

Methods / Tools: Python, GIS, JavaScript, Leaflet, Chart.js, ACS, HUD datasets, spatial joins, indicator construction, data visualization.

Key Contribution: Integrated multiple housing and demographic datasets, organized tract-level indicators, CBSA level indicators, and designed interactive mapping tools to support spatial equity analysis. Blog is published on the Housing Initiative at Penn.

Links: (This is an ongoing project and will be fully published soon)

For the blog: https://jyxu48.github.io/HCV-Dashboard_4.0/

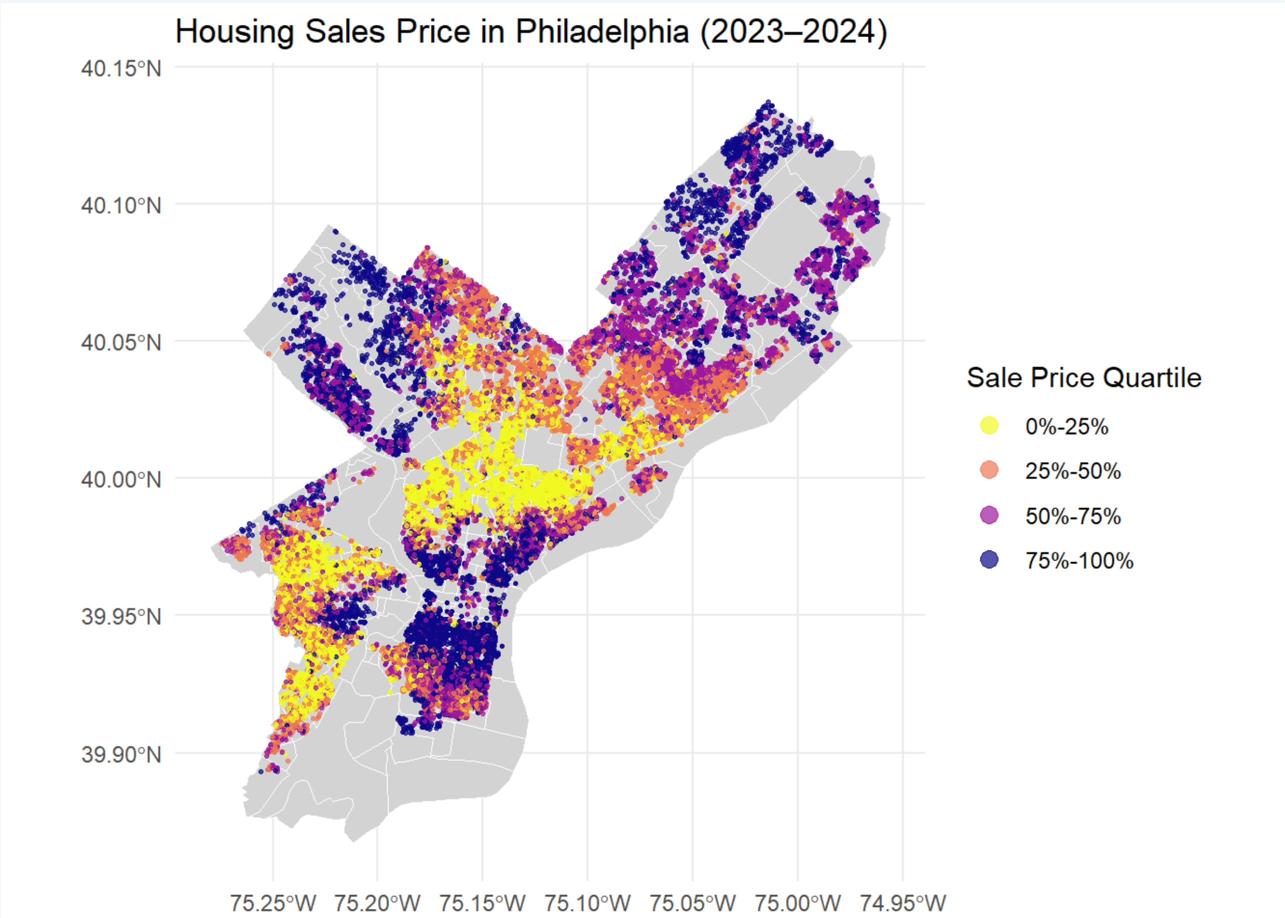
For the Interactive Map: https://drive.google.com/drive/folders/1wZh06ks3kDcQxYUd0GIRzEMFNd2PknFA?usp=sharing

For the Data Processing: (currently unavailable, please contact me if interested)

Interactive map can be downloaded through google drive and run on your local PC temporally. It will be held on a cloud server soon and can be visit through webpage. **Please Contact: jinyang48@outlook.com for more info**

2. Philadelphia Housing Price Modeling

Spatial housing market analysis and model comparison



model output map

Project Focus: Analyzed spatial variation in Philadelphia housing prices using statistical and spatial regression models.

Methods / Tools: Python or R, GIS, OLS, spatial lag model, geographically weighted regression, spatial visualization.

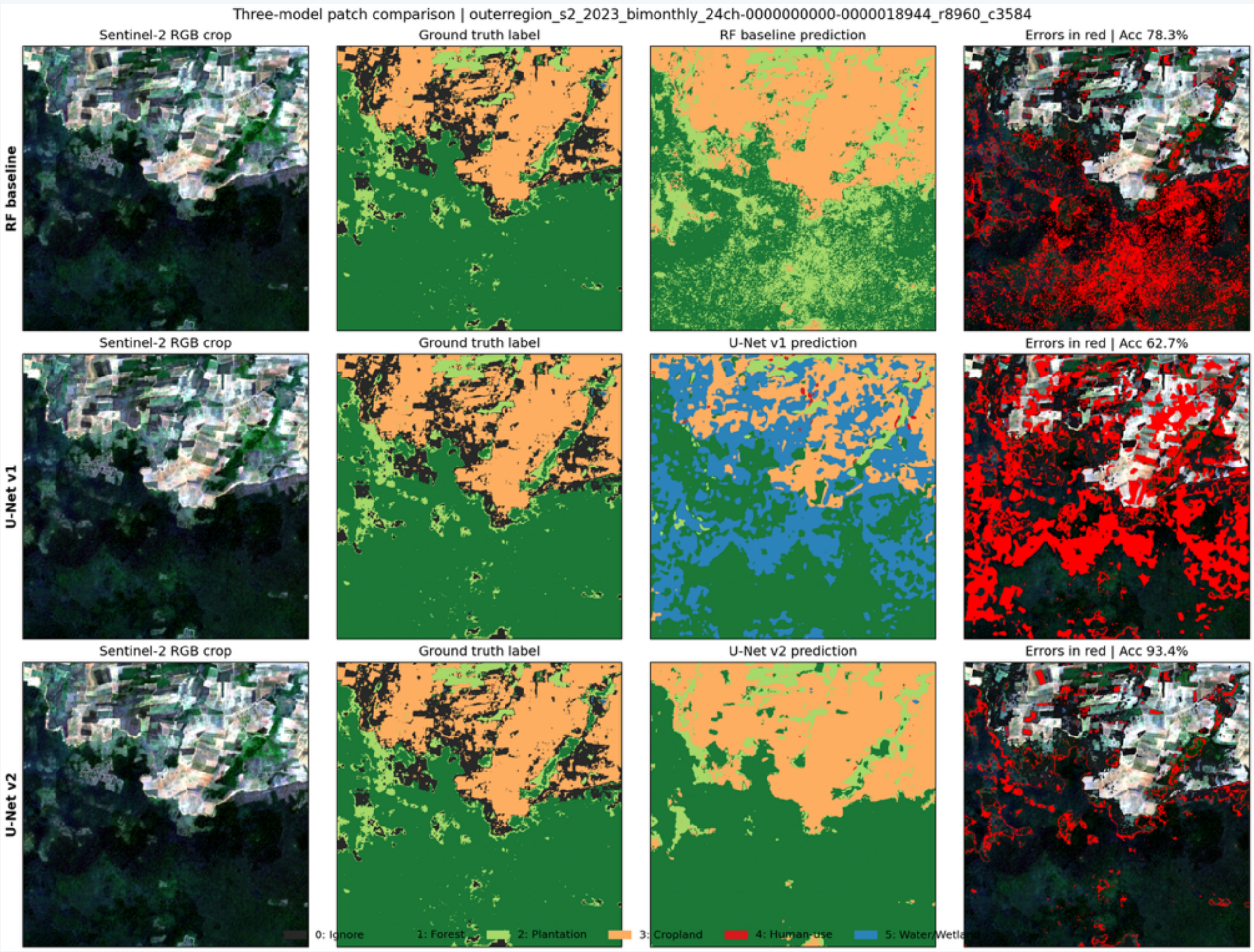
Key Contribution: Compared modeling approaches to evaluate how housing values vary across neighborhoods and how spatial relationships affect model performance.

Links:

Project technical appendix: https://jyxu48.github.io/Public-Policy_Analytics/Assignments/midterm/slides/Jinyang_Xu_appendix1.html

3. Satellite Deep Learning for Habitat Land-Cover Segmentation

Remote sensing, U-Net semantic segmentation, and environmental spatial analysis



Comparison of the outcome of the 3 models (RF, U-Net1 and U-Net2)

Project Focus: Developed a deep learning and remote sensing workflow to classify land-cover patterns related to Asian elephant habitat and human activity landscapes.

Methods / Tools: Satellite imagery, semantic segmentation, U-Net, Python, GIS, land-cover labels, model evaluation, mIoU, environmental data processing.

Key Contribution: Built an end-to-end workflow connecting remote sensing data preparation, land-cover class design, model training, prediction evaluation, and spatial interpretation.

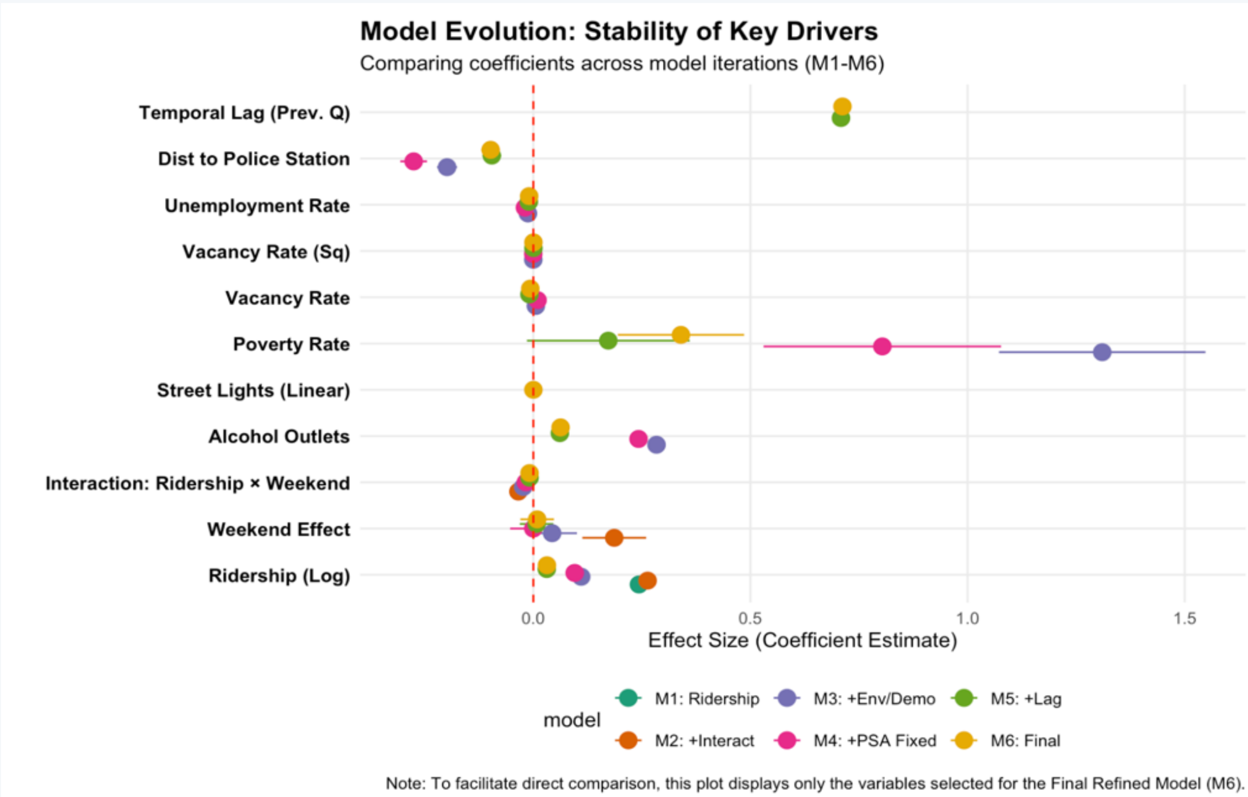
Links:

Project technical appendix: https://jyxu48.github.io/Satellite-Deep-Learning-project/

GitHub Repository: https://github.com/jyxu48/Satellite-Deep-Learning-project

4. Modeling Crime Risk Around SEPTA Bus Stops

Transit, public safety, neighborhood indicators, and spatial risk patterns



Key drivers of crime count around SEPTA bus stops

Project Focus: Investigated the relationship between transit ridership, neighborhood characteristics, and public safety conditions around SEPTA bus stops.

Methods / Tools: Python, GIS, spatial analysis, public safety data, transit stop data, neighborhood indicators, visualization.

Key Contribution: Built a spatial analysis workflow to identify risk patterns around transit infrastructure and support more evidence-based transportation planning.

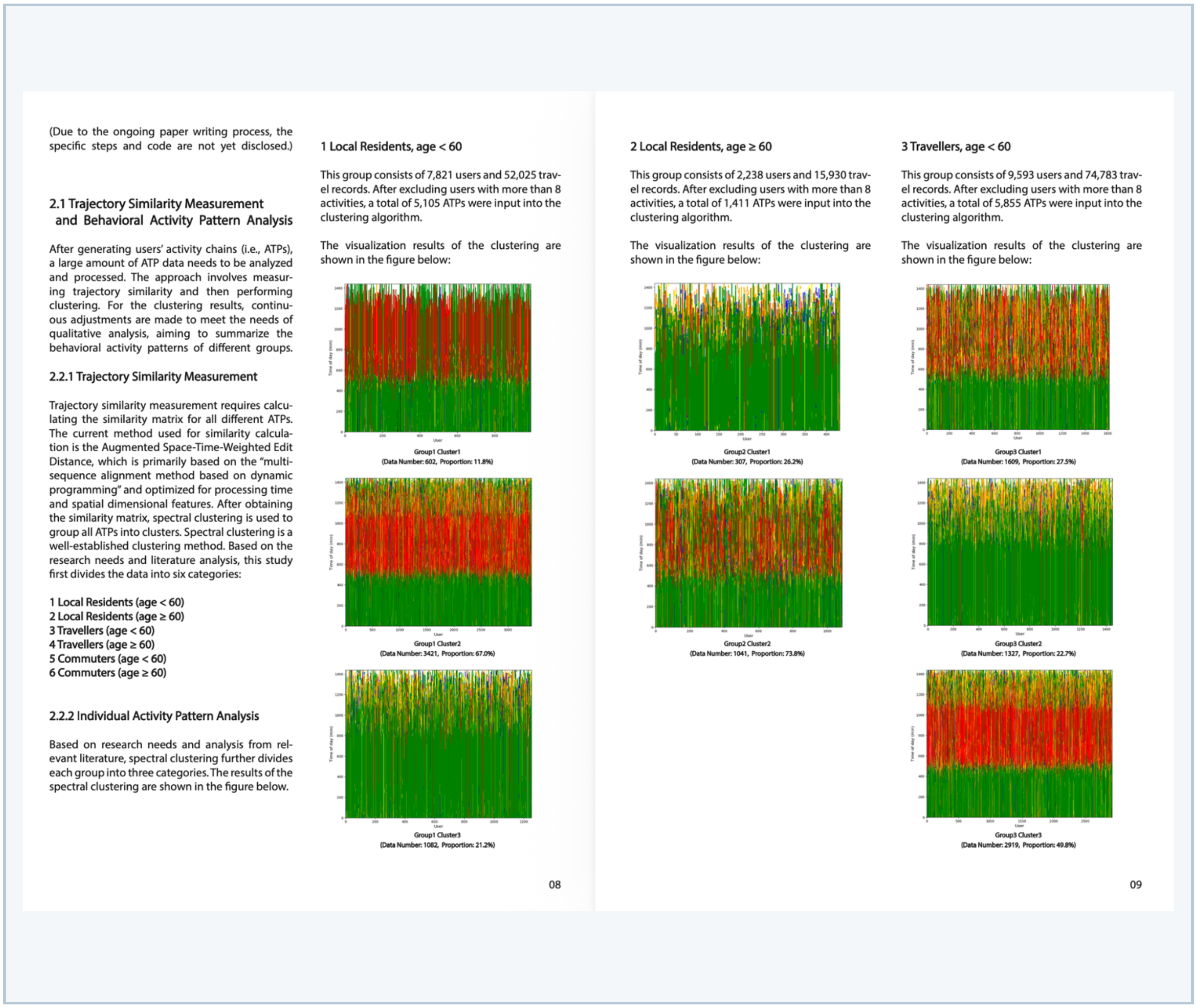
Links

Project technical appendix: https://jyxu48.github.io/Public-Policy_Analytics/Final/Final_appendix.html

Project slides: **https://jyxu48.github.io/Public-Policy_Analytics/Final/Slides/Slides/Simple_context.html#/title-slide**

5. Individual Activity Pattern Analysis Using Mobile Signal Data

Big data processing, activity chains, and urban mobility



(Due to the ongoing paper writing process, the specific steps and code are not yet disclosed.)

2.1 Trajectory Similarity Measurement and Behavioral Activity Pattern Analysis

After generating users' activity chains (i.e., ATPs), a large amount of ATP data needs to be analyzed and processed. The approach involves measuring trajectory similarity and then performing clustering. For the clustering results, continuous adjustments are made to meet the needs of qualitative analysis, aiming to summarize the behavioral activity patterns of different groups.

2.2.1 Trajectory Similarity Measurement

Trajectory similarity measurement requires calculating the similarity matrix for all different ATPs. The current method used for similarity calculation is the Augmented Space-Time-Weighted Edit Distance, which is primarily based on the "multi-sequence alignment method based on dynamic programming" and optimized for processing time and spatial dimensional features. After obtaining the similarity matrix, spectral clustering is used to group all ATPs into clusters. Spectral clustering is a well-established clustering method. Based on the research needs and literature analysis, this study first divides the data into six categories:

1 Local Residents (age < 60)
2 Local Residents (age ≥ 60)
3 Travellers (age < 60)
4 Travellers (age ≥ 60)
5 Commuters (age < 60)
6 Commuters (age ≥ 60)

2.2.2 Individual Activity Pattern Analysis

Based on research needs and analysis from relevant literature, spectral clustering further divides each group into three categories. The results of the spectral clustering are shown in the figure below.

1 Local Residents, age < 60

This group consists of 7,821 users and 52,025 travel records. After excluding users with more than 8 activities, a total of 5,105 ATPs were input into the clustering algorithm.

The visualization results of the clustering are shown in the figure below:

Group1 Cluster1
(Data Number: 602, Proportion: 11.8%)

Group1 Cluster2
(Data Number: 3421, Proportion: 67.0%)

Group1 Cluster3
(Data Number: 1082, Proportion: 21.2%)

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2 Local Residents, age ≥ 60

This group consists of 2,238 users and 15,930 travel records. After excluding users with more than 8 activities, a total of 1,411 ATPs were input into the clustering algorithm.

The visualization results of the clustering are shown in the figure below:

Group2 Cluster1
(Data Number: 307, Proportion: 26.2%)

Group2 Cluster2
(Data Number: 1041, Proportion: 73.8%)

3 Travellers, age < 60

This group consists of 9,593 users and 74,783 travel records. After excluding users with more than 8 activities, a total of 5,855 ATPs were input into the clustering algorithm.

The visualization results of the clustering are shown in the figure below:

Group3 Cluster1
(Data Number: 1609, Proportion: 27.5%)

Group3 Cluster2
(Data Number: 1327, Proportion: 22.7%)

Group3 Cluster3
(Data Number: 2919, Proportion: 49.8%)

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Project Focus: Analyzed individual activity patterns using mobile signal data and POI datasets in Nanjing.

Methods / Tools: Python, big data cleaning, POI matching, activity chain construction, sequence alignment, behavior pattern recognition, ArcGIS.

Key Contribution: Processed large-scale mobile signal data and developed analytical workflows to identify activity patterns and urban behavior structures.

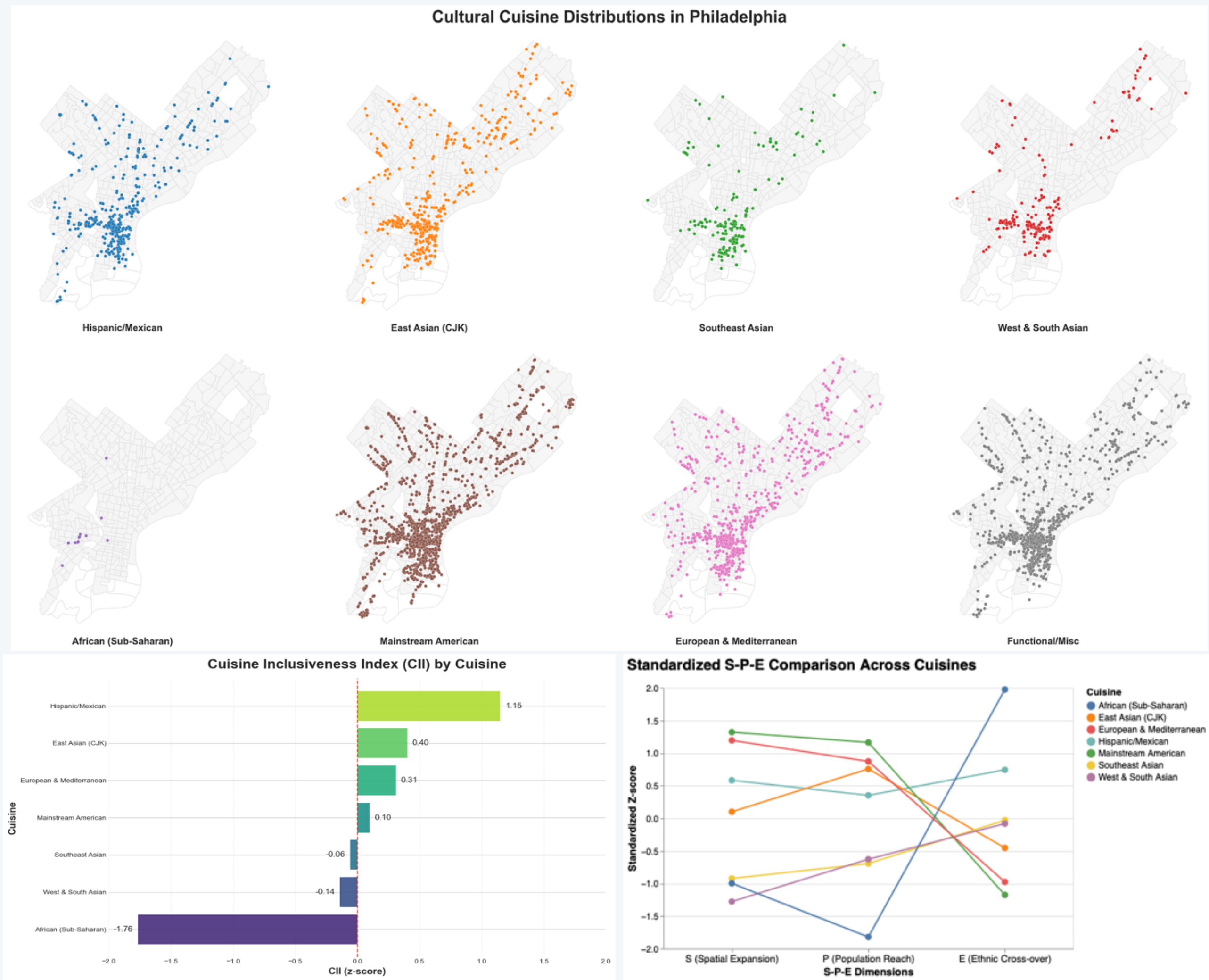
Relevance to GAI: Relevant to demographic behavior research, mobility analysis, urban trends, quantitative planning methods, and socioeconomic interpretation.

Links:

Work sample (PDF): https://jyxu48.github.io/JinyangXu_Website/Projects/mobile-signal-nanjing.pdf

6. Cuisine Influence Index (CII): Philadelphia Food Culture Diffusion

Python-based spatial analysis, demographic context, restaurant data, and cultural diffusion mapping



Project Focus: Built a web-based analytical project to understand how different cuisines travel across Philadelphia neighborhoods, asking which cuisines become visible across the city and which remain concentrated within ethnic enclaves.

Methods / Tools: Python, spatial joins, Yelp restaurant data, ACS demographic data, census tract boundaries, Shannon diversity, Moran's, and web-based documentation.

Key Contribution: Designed a Cuisine Influence Index (CII) combining Spatial Expansion, Population Reach, Ethnic Cross-over, and interaction effects to translate restaurant locations and neighborhood demographics into a measurable index of urban cultural diffusion.

Links:

Website: https://jyxu48.github.io/Jinyang_Xu_Python_Project_2025/

GitHub Repository: https://github.com/jyxu48/Jinyang_Xu_Python_Project_2025

7. LLM-Based Street-View Walkability Analysis

Street-view imagery, semantic segmentation, and walkability measurement on the 3 main street of Philly

LLM based street-view walkability score chart

Project Focus: Developed a computer-vision and LLM-assisted workflow to evaluate walkability conditions from Google Street View imagery.

Methods / Tools: Google Street View, semantic segmentation, LLM vision pipeline, Python, urban design indicators, visual interpretation.

Key Contribution: Translated visual urban environment features into measurable walkability indicators and structured outputs for urban analysis.

Links:

Project technical appendix: https://jyxu48.github.io/AI-for-Urban-Sustainability/